Because we're stuck on Earth, we're forced to peer at distant worlds through an ocean of distorting air. Much of the ultraviolet, infrared, and radio waves they emit do not penetrate our atmosphere. It's easy to see why our spacecraft have revolutionized the study of the Solar System: We ascend to stark clarity in the vacuum of space, and there approach our objectives, flying past them, as did *Voyager*, or orbiting them, or landing on their surfaces.

These spacecraft have returned four trillion bits of information to Earth, the equivalent of about 100,000 encyclopedia volumes. I described the *Voyagers 1* and *2* encounters with the Jupiter system in *Cosmos*. In the following pages, I'll say something about the Saturn, Uranus, and Neptune encounters.

JUST BEFORE *VOYAGER 2* was to encounter the Uranus system, the mission design had specified a final maneuver, a brief firing of the on-board propulsion system to position the spacecraft correctly so it could thread its way on a preset path among the hurtling moons. But the course correction proved unnecessary. The spacecraft was already within 200 kilometers of its designed trajectory—after a journey along an arcing path 5 billion kilometers long. This is roughly the equivalent of throwing a pin through the eye of a needle 50 kilometers away, or firing your rifle in Washington and hitting the bull's-eye in Dallas.

Mother lodes of planetary treasure were radioed back to Earth. But Earth is so far away that by the time the signal from Neptune was gathered in by radio telescopes on our planet, the received power was only 10^{-16} watts (fifteen zeros between the decimal point and the one). This weak signal bears the same proportion to the power emitted by an ordinary reading lamp as the diameter of an atom bears to the distance from the Earth to the Moon. It's like hearing an amoeba's footstep.

The mission was conceived during the late 1960s. It was first funded in 1972. But it was not approved in its final form (including the encounters with Uranus and Neptune) until after the ships had completed their reconnaissance of Jupiter. The two spacecraft were